

GEMP ACCESS COURSE: BIOLOGICAL CHEMISTRY

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Lecture 7: Alkynes

Kwame Nkrumah University of Science and Technology, Kumasi | Leaders In Change

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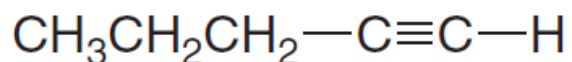
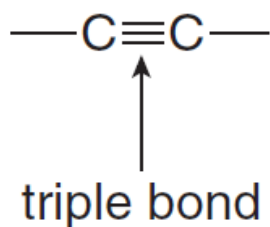
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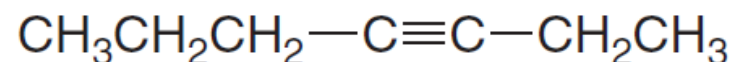
Alkyne

- **Alkynes** contain a carbon–carbon triple bond.
- **Terminal alkynes** has the triple bond at the end of the carbon chain, so that a hydrogen atom is directly bonded to a carbon atom of the triple bond.
- **Internal alkynes** has a carbon atom bonded to each carbon atom of the triple bond.

Alkyne



terminal alkyne

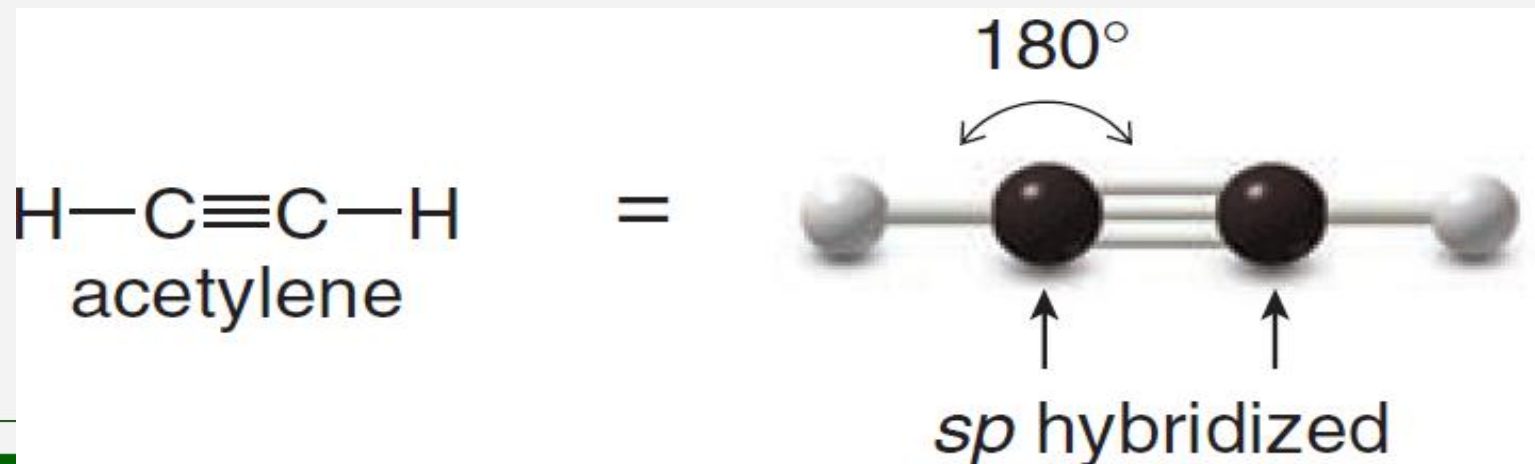


internal alkyne

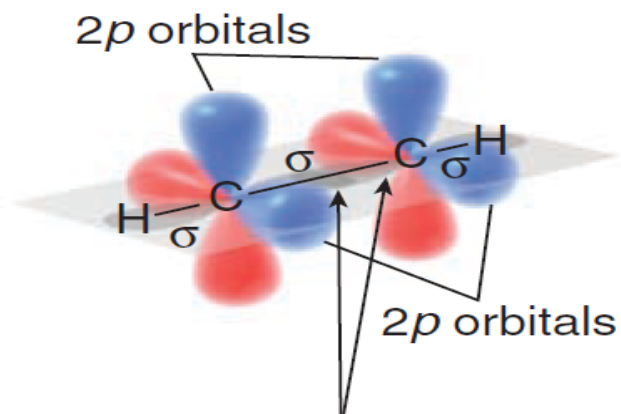


Alkyne

- A triple bond consists of a σ bond and two π bonds. The general formula for the alkynes is C_nH_{2n-2} . They are unsaturated and highly reactive compounds.
- Each carbon of a triple bond is sp hybridized and linear, and all bond angles are 180° .

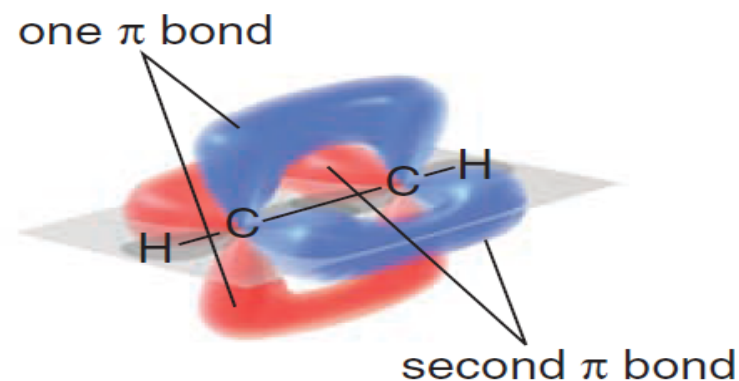


- Both π bonds of a C – C triple bond are weaker than a C – C σ bond, making them much more easily broken.
- As a result, alkynes undergo many addition reactions.
- Alkynes are more polarizable than alkenes because the electrons in their π bonds are more loosely held.



Overlap of the two sp hybrid orbitals forms the C–C σ bond.

Two π bonds extend out from the axis of the linear molecule.



Overlap of two sets of two $2p$ orbitals forms two C–C π bonds.



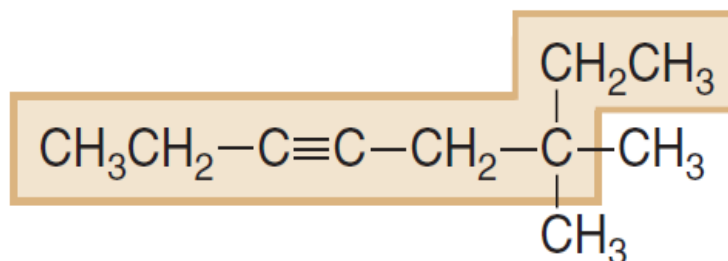
Nomenclature of Alkyne

- Alkynes are named in the same way that alkenes
- In the IUPAC system, change the **-ane** ending of the parent alkane to the suffix **-yne**.
- Choose the longest carbon chain that contains both atoms of the triple bond and number the chain to give the triple bond the lower number.



- Compounds with two triple bonds are named as **diynes**, those with three are named as **triynes**, and so forth.
- Compounds with both a double and a triple bond are named as **enynes**. The chain is numbered to give the first site of unsaturation (either C=C or C≡C) the lower number.

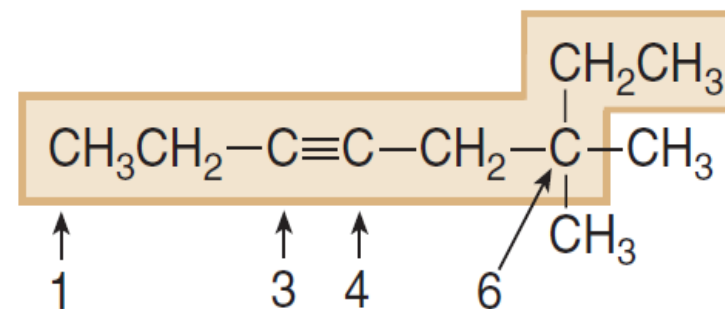
[1] Find the longest chain that contains both carbons of the triple bond.



8 C's in the longest chain

octane $\longrightarrow$ octyne

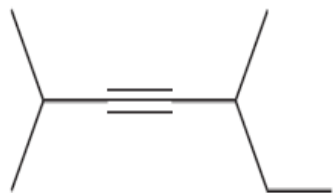
[2] Number the long chain; then name and number the substituents.



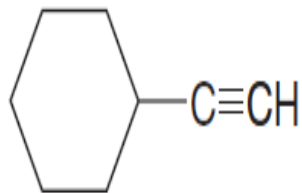
two methyl groups at C6

Answer: 6,6-dimethyl-3-octyne

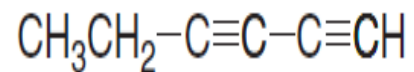




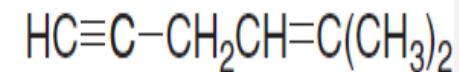
2,5-dimethyl-3-heptyne



ethynylcyclohexane



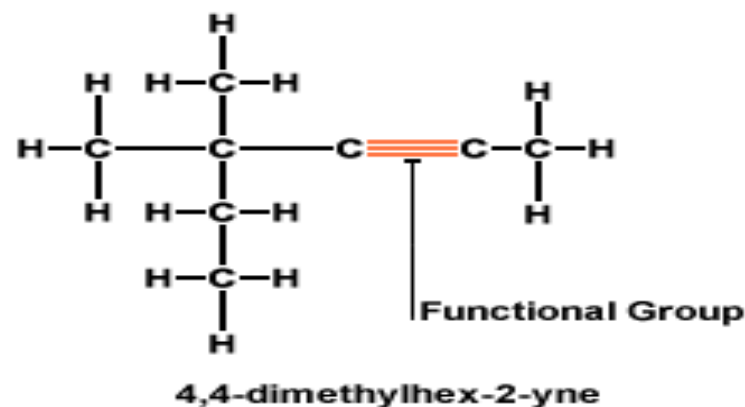
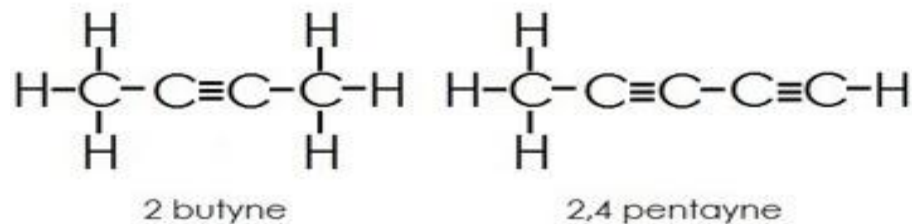
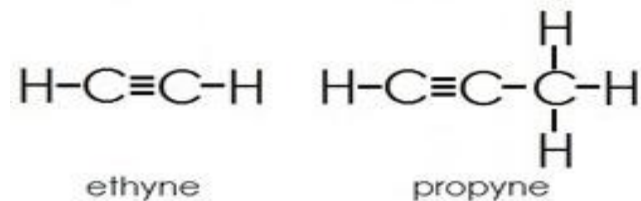
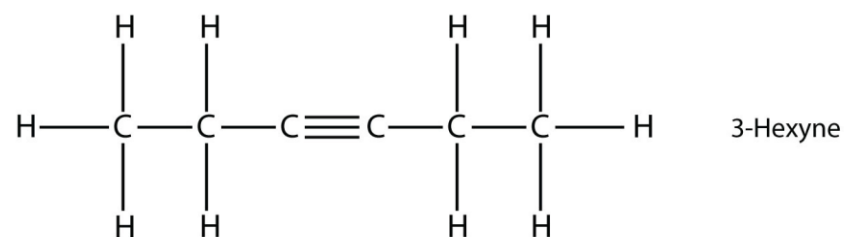
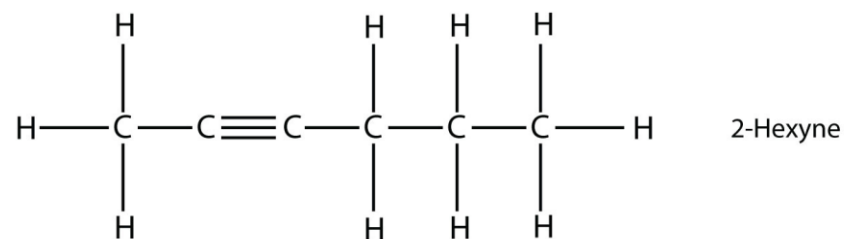
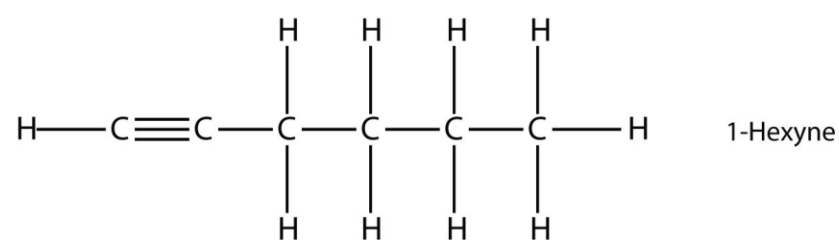
1,3-hexadiyne



5-methyl-4-hexen-1-yne

- The simplest alkyne, $\text{HC}\equiv\text{CH}$, named in the IUPAC system as **ethyne**, is more often called **acetylene**, its common name. The two-carbon alkyl group derived from acetylene is called an ethynyl group ($\text{HC}\equiv\text{C}-$).
- An important use of acetylene is in oxyacetylene gas welding. The fuel gas burns with oxygen in a torch. An incredibly high heat is produced, and this is enough to melt metal.

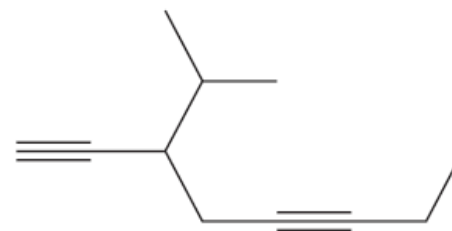




Give the IUPAC name for each compound.

- $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2\text{C}(\text{CH}_2\text{CH}_2\text{CH}_3)_3$
- $\text{CH}_3\text{C} \equiv \text{CC}(\text{CH}_3)\text{ClCH}_2\text{CH}_3$
- $\text{CH}_2=\text{CHCH}_2\text{CH}(\text{CH}_2\text{CH}_3)\text{C} \equiv \text{CC}(\text{CH}_3)_2\text{CH}_2\text{CH}_2\text{CH}_3$

d.



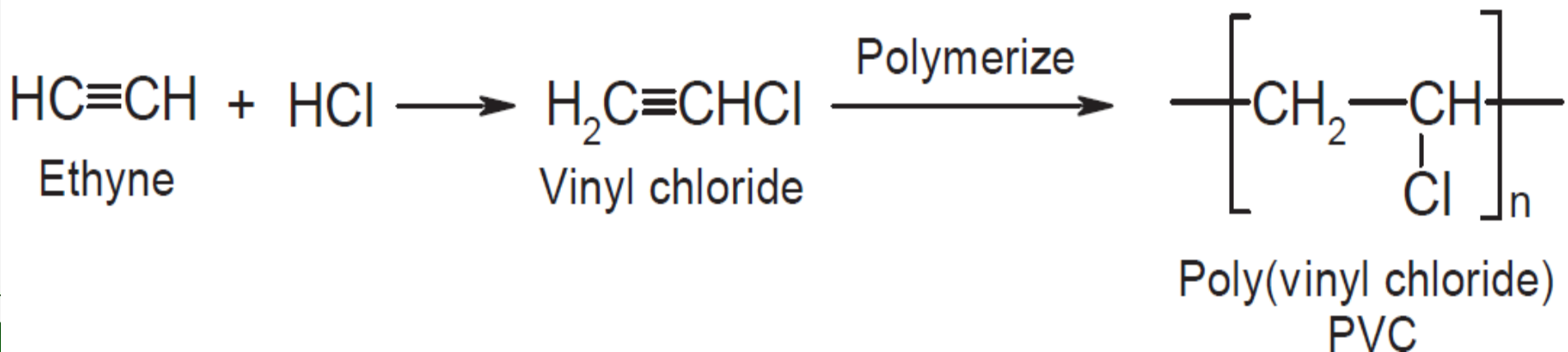
Give the structure corresponding to each of the following names.

- trans*-2-ethynylcyclopentanol
- 4-*tert*-butyl-5-decyne
- 3-methylcyclononyne



Industrial uses of alkynes

- ❑ Ethynes are industrially used as a starting material for polymers, e.g. plastic piping, Teflon and acrylics.
- ❑ **Polymers** are large molecules, which are prepared by linking many small monomers.
- ❑ Polyvinyl chloride, also commonly known as PVC, is a polymer produced from the polymerization of vinyl chloride.



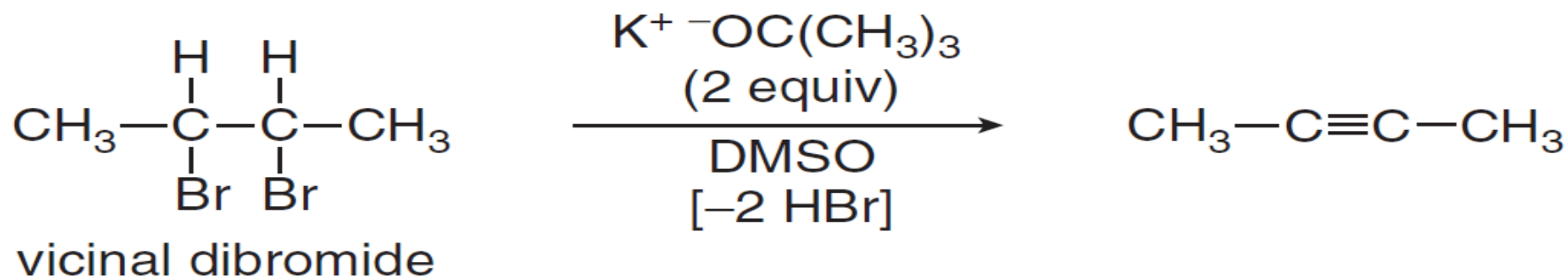
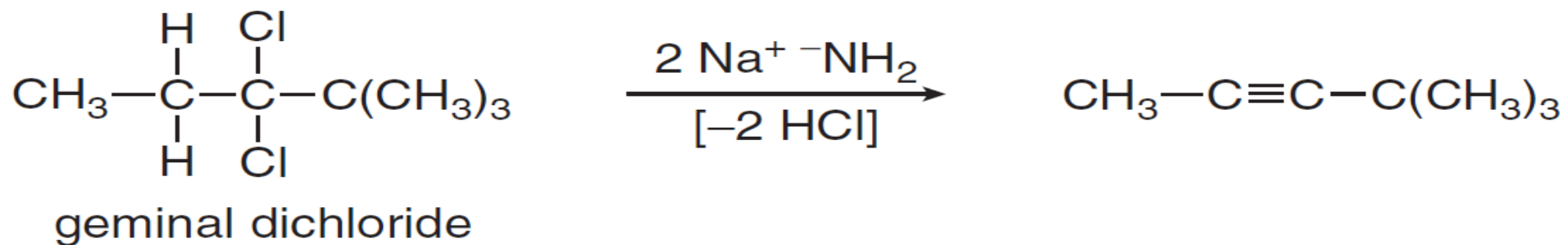
Physical properties of alkynes

- ❑• Alkynes have low melting points and boiling points.
- ❑• Melting points and boiling points increase as the number of carbons increases.
- ❑• Alkynes are soluble in organic solvents and insoluble in water because they have low polarity .



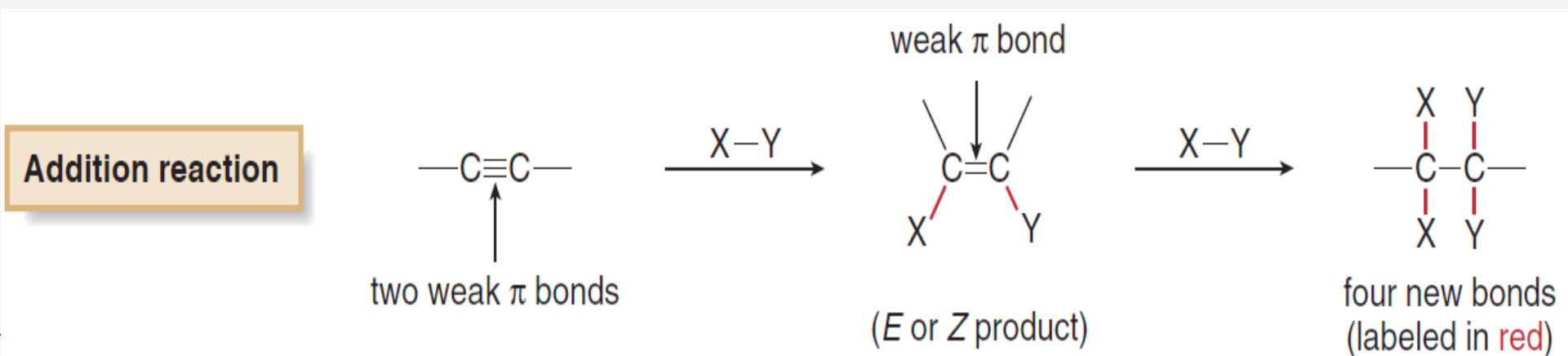
Preparation of alkynes

- Alkynes are prepared from alkyl dihalides via elimination of atoms or groups from adjacent carbons.
- A strong base removes two equivalents of HX from a vicinal or geminal dihalide to yield an alkyne

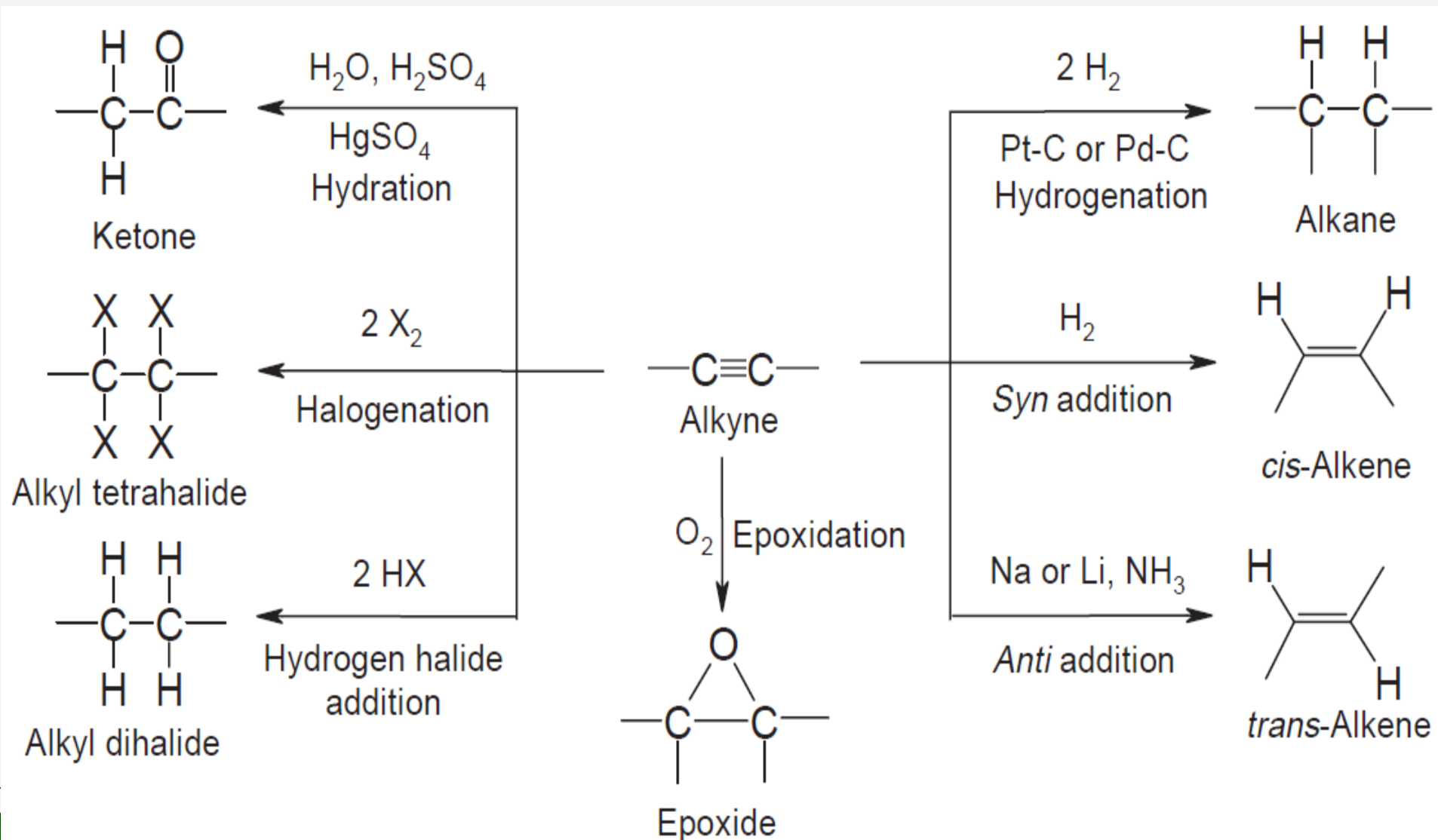


Reactions of Alkynes:

- ❖ Alkynes undergo addition reactions because they contain weak π bonds. Two sequential reactions take place.
- ❖ **Addition** of one equivalent of reagent forms an alkene, which then adds a second equivalent of reagent to yield a product having four new bonds.

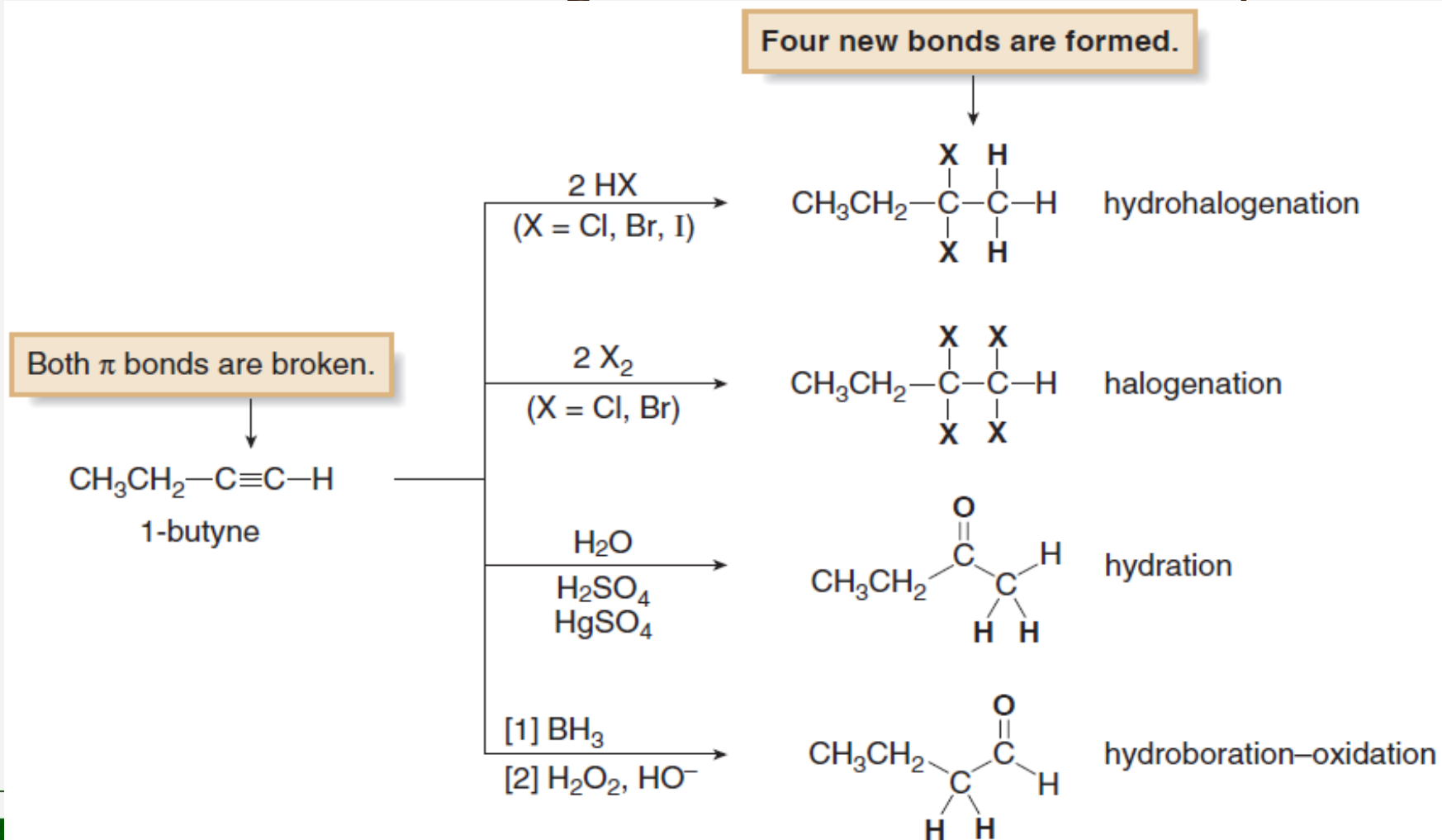


Reaction Map of Alkyne



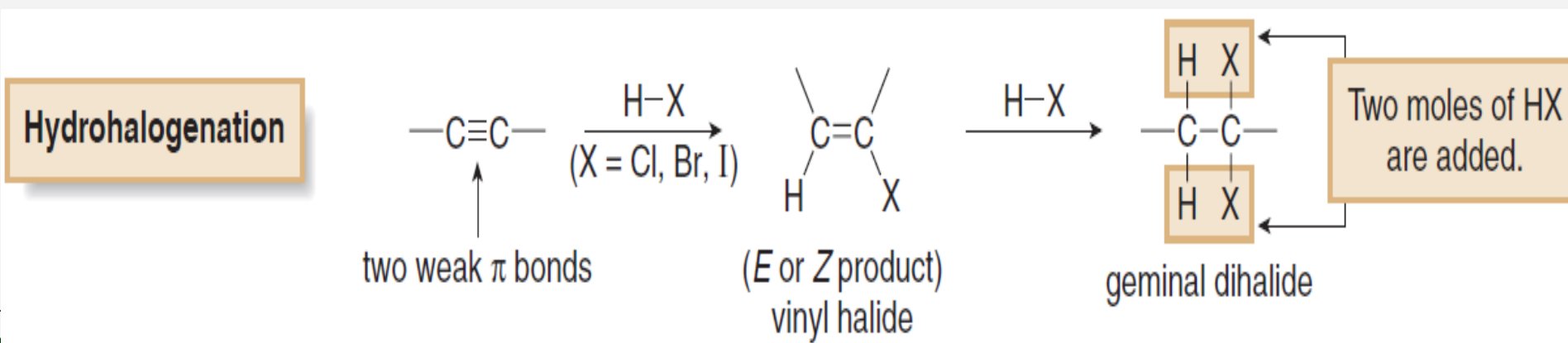
Addition Reactions

- ❖ Two molecules of reagent are needed for each triple bond.



1- Addition of Hydrogen Halides

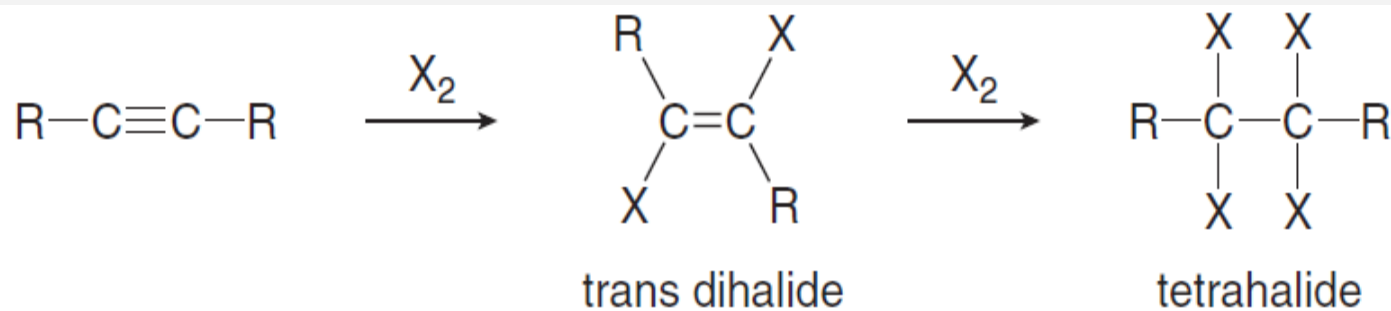
- ❑ Alkynes undergo hydrohalogenation, the addition of hydrogen halides, HX (X = Cl, Br, I).
- ❑ Two equivalents of HX are usually used: addition of one mole forms a vinyl halide, which then reacts with a second mole of HX to form a geminal dihalide.



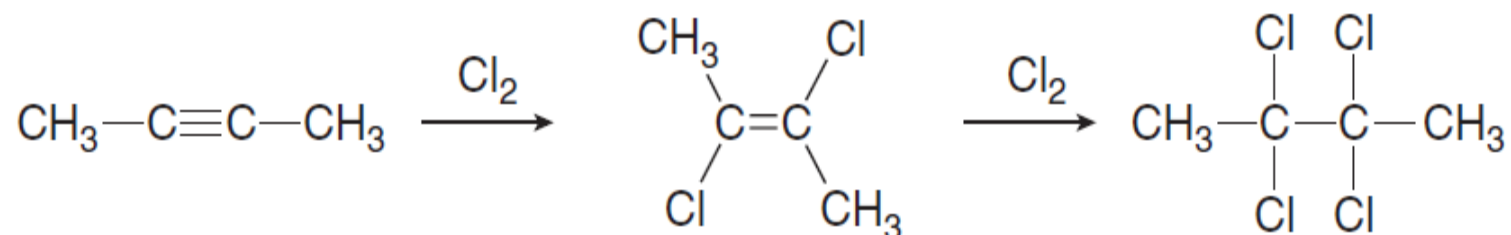
2- Addition of Halogen

□ Halogens, X_2 ($X = Cl$ or Br), add to alkynes in much the same way they add to alkenes. Addition of one mole of X_2 forms a trans dihalide, which can then react with a second mole of X_2 to yield a tetrahalide.

Halogenation

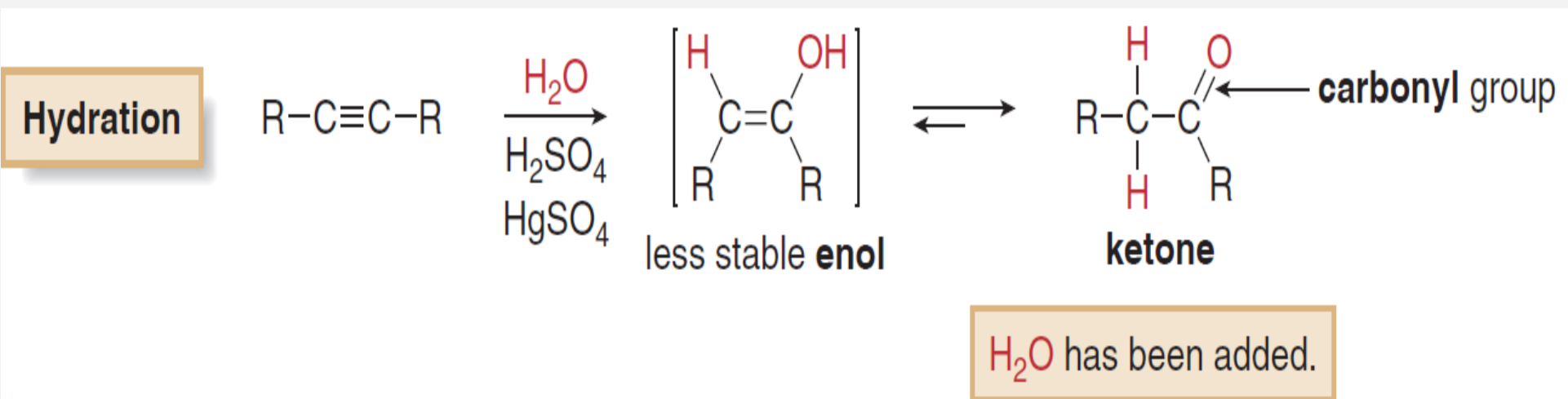


Example



3- Addition of Water

- In the presence of strong acid or Hg^{2+} catalyst, the elements of H_2O add to the triple bond, but the initial addition product, an **enol**, is unstable and rearranges to a product containing a carbonyl group that is, a $\text{C}=\text{O}$. A carbonyl compound having two alkyl groups bonded to the $\text{C}=\text{O}$ carbon is called a ketone.



4- Addition of hydrogen

- ❑ A catalysts such as **palladium** is normally needed for this reaction.

